

Sruti Bhattacharjee

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Education

KTH Royal Institute of Technology

MSc, Interactive Media Technology

Thesis: Redesigning No-Code Virtual Reality Platform for K12 Education with Teachers' Tacit Knowledge

Sweden

Ongoing

École Polytechnique Fédérale de Lausanne (EPFL)

Graduate Study Abroad, School of Computer and Communication Sciences

Switzerland

2021 - 2023

West Bengal University of Technology

BTech, Information Technology

India

2019

Indian Institute of Technology Kharagpur

Undergraduate Visiting Student, Department of Computer Science and Engineering

India

2017 - 2018

Thesis: Counterfeit Integrated Circuit Detection using Computer Vision and Image Processing Techniques

Relevant Experience

ETS (Educational Testing Service) — through Jyväskylä Yliopisto

Finland

Accessible Graph Navigation for Blind Users

Aug 2025 - Aug 2025

- Uncovered digital accessibility gap for blind users, defining a real-time graph navigation concept for solo & group workflows
- Translated blind-user pain points into phygital product concept, connecting interaction needs with graph-navigation use cases
- Prototyped graph sonification using Python, translating visual heuristics to non-visual interactive navigation features for MVP
- Prioritized sonification over haptic & thermal features using Eisenhower Decision Principles within a 7-day delivery constraint

Jupyter Thebe — through University of Lausanne

Switzerland

Assistive Technology for Hybrid Programming Classes

Dec 2022 - Feb 2023

- Identified adoption friction in Thebe (a Programming assistant chatbot for Jupyter Notebook) by mapping gaps between the intended product use cases and novice student needs by conducting semi-structured interviews with 38 sophomore students
- Generated 24 student-driven UI concepts by deploying a participatory workshop across 8 hybridly located student groups of 4
- Defined 10 actionable product requirements by translating UI insights into user-centric features for hybrid group & solo work

EPFL

Switzerland

Emotion Recognition from Facial Expression in Virtual Reality

Sep 2021 – Feb 2022

- Co-developed a real-time emotion-recognition pipeline in VR, reaching 45\% accuracy in LightGBM and 41% in Random Forest
- Developed a Unity3D-based prototype following FACS (Facial Action Coding System) for avatar facial expression data collection
- Collected data of 27 subjects & developed a database of avatar blendshapes \& raw lips using HTC Vibe HMD & facial tracking

EPFL

Switzerland

Phygital Civic Engagement Platform with AI & API

Sep 2021 - Feb 2022

- Transformed collaborative city activism by proposing a phygital platform with colleagues (2 Architects, 3 computer engineers)
- Activated underutilized physical resources by connecting idle supply with active user demand by designing an AI-integrated recommendation engine based on semantic analysis of user activity, interests, and location, with API-based feedback loops

Ericsson

Sweden

Haptic Perception in Teleoperation Robotic Systems

Jun 2021 - Sep 2021

- Evaluated haptic perception during network changes while remotely pushing physical objects in robotic teleoperation system
- Led experiment design and front-end development for a delivered prototype that uses a modular LEGO object to manipulate physical load, eliciting the impact of network variables (latency, jitter, camera constraints) on haptic perception via A/B testing
- Developed a teleoperation system connecting 2 Omni haptic pens & Ericsson internal network server with a backend engineer

Intel — through IIT Kharagpur

India

DeepFakeFace Video Detection

Jan 2020 – Mar 2020

- Built a pipeline for original vs. deepfakeface video detection based on wrinkle movement by analyzing inter-frame differences

Mentoring Experience

Teaching Assistant (UG & PG) — KTH Royal Institute of Technology

Sep 2020 - Jun 2025

- Machine Learning • Programming Fundamentals • Multivariable Calculus • Computer Networking • Visualisation

Management Consulting Case Mentor (UG & PG) — EPFL

Aug 2022 - Jan 2023

- EPFL Consulting Society: Mentored students of EPFL, UNIL, HEC Lausanne of diverse backgrounds in weekly studies for 1 semester

Visiting Lecturer (UG) — NSHM Knowledge Campus, Kolkata

Aug 2019 - Dec 2019

- Interactive 3D Computer Graphics using OpenGL and other Languages (WBUT AICTE and Non-AICTE syllabus followed)

Volunteer Experience

HackZurich

Coordinated logistics & event operations for 500+ developers during the 47-hours AI hackathon

Apr 2021 - Sep 2021

KTH AI Society

Conducted 5 seminars with Guest Lecturers, 1 hackathon & initiated the AI Podcast (3 interviews)

Oct 2020 - Aug 2021

Skills

- **UX Research:** Diary Study, A/B Testing, Soma Design, Contextual Inquiry, Survey, Interviewing, Design Fiction, Co-design Workshop
- **Programming & Tools:** Klaxoon, Figma, Miro, Unity3D, Python, PyTorch, TensorFlow, OpenCV, GLSL, Plotly, Jax, Git, Jira